

FinRelief AI — Demo Script (Final Version)

6–7 Minutes | Simple English

SECTION 1 — INTRODUCTION (0:00 – 0:35)

[Show: Browser closed, just desktop visible]

"Hello everyone. Today I will show my project — **FinRelief AI**.

The problem is — many people in India have multiple loans. They don't know how to talk to banks. They don't know how much to offer for settlement. And they don't know their legal rights.

So I built **FinRelief AI** — a website where a borrower can:

- Add and manage all their loans in one place
- View the details of each loan
- Get an AI-powered settlement plan
- Generate a professional letter to send to the lender
- Know their legal rights as a borrower

Everything is personal — the AI reads YOUR financial data and gives YOU a specific plan.

Let me first explain the technologies, then I will show the full website."

SECTION 2 — TECH STACK (0:35 – 1:10)

[Show: Open `E:\project\requirements.txt` in VS Code]

"Here is my `requirements.txt` — this shows all the backend libraries.

For the **backend server** I used **FastAPI** — a Python framework. Very fast and modern.

For the **database** I used **SQLite** with **SQLAlchemy** — this stores all user data, loans, and history.

For **login security** I used **JWT tokens**. When you log in, the server creates a token. Every page sends this token — so only you can see your own data. Passwords are never stored directly — they are **hashed** for security.

For **AI** — I connected **Google Gemini 2.5 Flash**. It reads the loan data and writes the strategies and letters.

For the **frontend** I used **React.js with Vite** — and **Axios** to connect the frontend to the backend.

That's the full tech stack. Now let me show you the website."



UI EXPLORATION — ALL PAGES

SECTION 3 — LOGIN PAGE (1:10 – 1:35)

[Show: Open browser → `http://localhost:3001`]

"This is the **Login Page**. Dark professional design. A borrower can register a new account or log in here.

I will log in with our test user — Fathima Noor."

[Action: Type email and password → click Login]

"Perfect — logged in. The system creates a JWT token and saves it. Every page from now on uses this token automatically, so I stay logged in everywhere."

SECTION 4 — DASHBOARD (1:35 – 2:25)

[Show: Dashboard page]

"This is the **Dashboard** — the main page.

At the top you can see the user's key numbers:

- Total loans and total outstanding amount
- Financial health score
- Debt stress level — it shows Low, Medium, or High based on how much of the income goes to EMI payments
- How many settlement records and AI history entries exist

Below that is the **Loan Priority List**. Every loan is ranked — High, Medium, or Low priority — based on the interest rate and how many months it is overdue. So the user knows which loan to handle first.

All this data is calculated fresh every time you open the dashboard."

SECTION 5 — LOANS (ADD & VIEW) (2:25 – 3:05)

[Show: Click on Loans section or loan list on Dashboard — wherever loans are visible]

"This is where the user can **manage all their loans**.

You can add a new loan by entering:

- Loan type — Personal Loan, Home Loan, Car Loan, etc.
- Lender name
- Total loan amount and outstanding balance
- Interest rate and EMI
- Due date and how many months overdue

Once added, the loan appears in the list. You can view its full details — outstanding amount, interest rate, risk category, recommended settlement amount — everything in one place.

All loans are linked to your account only. Nobody else can see your loans."

SECTION 6 — SETTLEMENT PREDICTOR (3:05 – 3:45)

[Show: Click "Settlement Predictor" in navigation]

"This is the **Settlement Predictor** — the most important feature.

The user enters their income and expenses. Then the system looks at all their loans and calculates — for each loan — how much to offer the bank for a one-time settlement.

The recommended amount is usually between **40% to 75%** of what you owe — depending on the risk level of that loan.

Then Google Gemini AI reads all this and gives a full analysis — which loan is most urgent, what actions to take, and how much total savings you can make by settling instead of paying in full.

Look at the result — each loan has a risk category, recommended settlement amount, and AI advice."

SECTION 7 — NEGOTIATION EMAIL (3:45 – 4:25)

[Show: Click "Negotiation Email" in navigation]

"This is the **Negotiation Email Generator**.

The user picks a loan, enters their details, and the system asks Google Gemini AI to write a **complete formal letter** to the bank.

The letter includes:

- Borrower's name and hardship situation
- The settlement amount being offered and why
- A request to remove penalty interest
- A request for a No Objection Certificate after payment

This is a letter a borrower would normally pay a lawyer to write. Here it is generated in seconds, personalized for their specific loan and lender."

[Show: Generated letter on screen]

"You can read the full letter here — completely ready to send."

SECTION 8 — KNOW YOUR RIGHTS (4:25 – 4:50)

[Show: Click "Know Your Rights" in navigation]

"This is the **Know Your Rights** page.

Many borrowers get harassed by recovery agents and don't know what is legal and what is not. This page explains:

- RBI rules for fair recovery
- What agents are NOT allowed to do
- How to file a complaint with the Banking Ombudsman

Knowledge is power — this page gives borrowers the confidence to stand up for themselves."

SECTION 9 — AI HISTORY (4:50 – 5:10)

[Show: Click "History" in navigation]

"This is the **History** page.

Every AI strategy and letter the user generates is saved here automatically. So you can always come back and read your old results.

The history is private — you only see your own records, nobody else's."



CODE WALKTHROUGH

SECTION 10 — CODE TOUR (5:10 – 6:20)

[Open VS Code — keep browser minimized for this whole section]

Step 1 → Show `E:\project\app\main.py`

"This is `main.py` — the starting point of the backend. It registers all 8 route modules and sets up CORS so the frontend can talk to the backend. Version 2.0 of the API."

Step 2 → Show `E:\project\app\core\logic.py` → scroll to line 100

"This is `logic.py`. This is where financial calculations happen — EMI ratio, stress level, surplus, settlement percentages. Pure Python logic, no AI needed here."

Step 3 → Show `E:\project\app\core\ai_engine.py` → scroll to line 354

"This is `ai_engine.py` — the AI brain of the project. This function builds a prompt with all the loan data and sends it to Google Gemini. If Gemini is not available or the rate limit is hit, it automatically falls back to a rule-based template so the user always gets an output."

Step 4 → Show `E:\project\app\routers\auth.py`

"This is `auth.py`. This handles login and registration. It hashes passwords using Werkzeug, checks credentials, and creates JWT tokens using HS256 algorithm that expire after 24 hours."

Step 5 → Show `E:\project\app\routers\ai_history.py` → line 71

"This is `ai_history.py`. Here you can see the history endpoint — it

filters by `current_user.user_id` so you only get your own data. This pattern is used across all endpoints for security."

Step 6 → Open browser → `http://localhost:8001/docs`

"Finally — FastAPI generates this interactive API documentation automatically. All 8 modules, all endpoints, all clearly listed. Protected endpoints need the JWT token. This is one of the big advantages of FastAPI."

SECTION 11 — CLOSING (6:20 – 6:45)

[Show: Switch back to Dashboard in browser]

"So to summarize — **FinRelief AI** is a complete web application built to help borrowers manage debt smartly.

7 pages on the frontend. 8 backend modules. JWT security. SQLite database. Google Gemini AI for personalized content.

The user can manage loans, predict settlements, generate negotiation letters, and know their rights — all in one place.

Thank you for watching."

✓ CHECKLIST BEFORE RECORDING

- ☐ Backend running: `.\venv\Scripts\uvicorn.exe app.main:app --host 127.0.0.1 --port 8001`
- ☐ Frontend running: inside `E:\project\frontend` → `npm run dev`
- ☐ Browser open at `http://localhost:3001` — logged OUT so you can show login
- ☐ VS Code open with `E:\project` folder loaded

- ☐ Practice the login → dashboard → loans → settlement → email → rights → history flow once before recording

CODE FILES ORDER (Section 10)

Order	File	What to say
1	app\main.py	Entry point, 8 modules, version 2.0
2	app\core\logic.py line ~100	Financial calculations, stress level
3	app\core\ai_engine.py line ~354	Gemini prompt, fallback logic
4	app\routers\auth.py	Password hash, JWT token
5	app\routers\ai_history.py line ~71	user_id filter, privacy
6	http://localhost:8001/docs	Auto API docs